

# Data-Driven Cognitive Phenotyping in Acquired Brain Injury

## A Clustering Analysis with Imputation Robustness Assessment

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# Outline

Introduction

Data & Methods

Results

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Conclusions

# Acquired Brain Injury: The Clinical Challenge

- **ABI** includes TBI, stroke, hypoxic injury, encephalitis
- Leading cause of long-term disability in working-age adults
- Cognitive effects span attention, memory, executive function, language, visuoperception
- **Key problem:** similar injury severity → very different cognitive profiles

## Why it matters

Severity bands and diagnostic labels group patients with substantively different cognitive profiles together, and the within-group variation that could guide treatment is hidden.

# Research Questions & Hypotheses

## Research Questions

1. Does clustering find **distinct cognitive phenotypes**?
2. Are phenotypes **stable** across imputation methods?
3. Does cluster membership **link to clinical unit**?
4. Does **domain-level** clustering beat variable-level?

## Formal Hypotheses

- H1  $\geq 2$  phenotypes, Silhouette  $> 0.40$ , noise  $< 30\%$
- H2 Pairwise ARI  $> 0.50$  across imputation methods
- H3  $\chi^2$  significant, Cramér's  $V > 0.10$
- H4 Domain wins all 3 metrics: Silhouette, VIF, cond. number

## Dataset Overview

- Raw extract: **22,075 assessments, 8,739 patients**
- Single neurorehabilitation centre
- 31 original columns (25 cognitive + 6 admin)
- Routine clinical data
- Large & systematic missingness

**Tier 1 Filtering** (50% threshold +  $\geq 1$  cognitive test):

- **14 eligible variables, 6 cognitive domains**
- **17,406 assessments from 7,285 patients**

**Complete-case analysis:**

- Only **9,245 rows (53.1%)** have all 14 tests

| Domain             | Vars      |
|--------------------|-----------|
| Orientation        | 3         |
| Attention          | 2         |
| Visuoperception    | 1         |
| Language           | 3         |
| Memory             | 4         |
| Executive Function | 1         |
| <b>Total</b>       | <b>14</b> |

## Imputation Methods: A 5-Category Taxonomy

| Category                 | Methods                                        |
|--------------------------|------------------------------------------------|
| Conventional Statistical | Mean, EM, MICE                                 |
| Machine Learning         | KNN, MissForest                                |
| Hybrid                   | Predictive Mean Matching (PMM)                 |
| Matrix Completion        | SoftImpute, NMF                                |
| Deep Learning            | Denoising Autoencoder, Variational Autoencoder |

- Extends Afkanpour et al. (2024) 4-category taxonomy → adds **Deep Learning**
- DAE: 15→128→64→32→64→128→15, **pattern-aware** corruption 0.2, 5 runs averaged
- VAE: same architecture, latent dim = 32, **cyclical**  $\beta$ : 0 → 0.1,  $K = 20$  samples averaged
- **10 methods** × **2 representations** = 20 pipeline runs

# Deep Learning Improvements

## DAE: Pattern-Aware Corruption

- Standard: corrupt random 20% of values
- Problem: real missingness is **structured** (tests given together)
- Solution: 50% of corruption masks sampled from empirical co-occurrence patterns
- Result: DAE learns from realistic missing patterns

## VAE: Cyclical $\beta$ -Annealing

- Standard: fixed  $\beta = 0.1$  on KL term
- Problem: **posterior collapse** — latent space ignored early in training
- Solution:  $\beta$  ramps  $0 \rightarrow 0.1$  over 40-epoch cycles
- Result: better latent space + reconstruction

**Held-out RMSE:** 10% holdout evaluation confirms DAE outperforms VAE on all variables

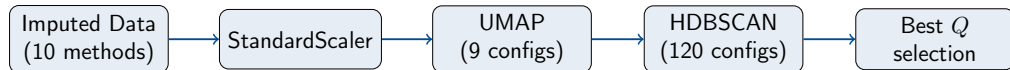
## Method Comparison at a Glance

| Category         | Method     | Year | Sil.  | $k$ | Key Strength              |
|------------------|------------|------|-------|-----|---------------------------|
| Conv. Stat.      | Mean       | —    | 0.604 | 4   | Simplest baseline         |
|                  | EM         | 1977 | 0.587 | 4   | MLE under normality       |
|                  | MICE       | 2011 | 0.598 | 3   | Gold standard; flexible   |
| Machine Learning | KNN        | 2001 | 0.576 | 4   | Non-parametric; local     |
|                  | MissForest | 2012 | 0.606 | 3   | Non-linear; mixed types   |
| Hybrid           | PMM        | —    | 0.498 | 4   | Preserves distribution    |
| Matrix Compl.    | SoftImpute | 2010 | 0.571 | 4   | Efficient; regularised    |
|                  | NMF        | 1999 | 0.546 | 3   | Non-negativity natural    |
| Deep Learning    | DAE        | 2018 | 0.510 | 5   | Deep non-linear           |
|                  | VAE        | 2014 | 0.603 | 4   | Probabilistic uncertainty |

Methods span 1977–2018.  $k$  = clusters from HDBSCAN; Sil. = silhouette score.



# Analysis Pipeline: UMAP + HDBSCAN



## UMAP grid:

- `n_neighbors`: 15, 30, 50
- `min_dist`: 0.0
- `n_components`: 3, 5, 8

## HDBSCAN grid:

- `min_cluster_size`: 500–3000
- `min_samples`: 5, 10, 25, 50
- `method`: eom, leaf
- `epsilon`: 0.0, 0.1, 0.5

$$Q = \text{Silhouette} \times (1 - \text{noise proportion})$$

+ Reliability-weighted domains, RobustScaler, consensus UMAP (10 seeds), KNN noise reassignment

# Shared Pipeline Design for H2

## Problem

Many moving parts vary across methods: scaling, UMAP embedding, cluster boundaries.

**Solution — isolate the imputation effect:**

1. Fit StandardScaler on **MICE** → `transform()` all others
2. Fit UMAP on **MICE** → `transform()` all others
3. Fit  $K$ -Means on **MICE embedding** → `predict()` all others
  - Same  $k$ , same boundaries, same embedding geometry
  - **Only source of variation:** the imputed values themselves
  - HDBSCAN labels retained for H1/H3/H4;  $K$ -Means labels for H2

## Clustering Results Across Imputation Methods

| Method     | Clusters | Silhouette | Noise % | Clustered $N$ |
|------------|----------|------------|---------|---------------|
| Mean       | 4        | 0.604      | 3.3     | 16,832        |
| KNN        | 4        | 0.576      | 7.3     | 16,135        |
| MICE       | 3        | 0.598      | 7.8     | 16,048        |
| MissForest | 3        | 0.606      | 14.4    | 14,899        |
| PMM        | 4        | 0.498      | 4.1     | 16,692        |
| EM         | 4        | 0.587      | 7.4     | 16,118        |
| SoftImpute | 4        | 0.571      | 16.9    | 14,464        |
| NMF        | 3        | 0.546      | 12.4    | 15,248        |
| DAE        | 5        | 0.510      | 4.3     | 16,657        |
| VAE        | 4        | 0.603      | 16.6    | 14,516        |

- Cluster count ranges from 3 to 5 across methods (primary MICE: 3 tiers)
- Silhouette on the HDBSCAN core (pre-reassignment); the final full-coverage MICE solution scores **0.526**
- Best UMAP: `n_components=8`; HDBSCAN: `min_cluster_size=1000`, `min_samples=50`, `eom`

## H1: Discrete, Stable Clusters — Supported

### Per-method bootstrap (50 iters):

- 10 / 10 methods clear pre-registered thresholds  
(median silhouette  $> 0.40$ , pre-KNN noise  $< 0.30$ )
- Cluster count 3–5 across methods (MICE: 3)

### Per-cluster core Jaccard (Hennig):

- Above-Average: **0.63**, Global Impairment: **0.62**
- Near-Normal (middle band): 0.47

**Three severity tiers** (a severity gradient, not dissociated profiles):

**Above-Average** ( $n = 6,725$ ): above cohort mean on all domains

### Verdict

#### Supported

10 / 10 methods clear sil + noise thresholds (pre-registered:  $\geq 8/10$ )

PC1 = 56% of variance: the cohort sits on a single overall-severity axis.

## H2: Imputation Robustness — Supported

*ARI is chance-corrected concordance: 0 = random, 1 = perfect. 0.50 = substantial overlap; 0.90+ = near-identical clusterings.*

### Shared pipeline:

- Mean pairwise ARI: **0.710**  
*95% bootstrap CI [0.704, 0.716]*
- Every one of the 45 pairs exceeds 0.50 (range 0.51–0.88)
- Consensus confidence: **0.929**; high-confidence assessments: **86.2%**

### Holm-corrected pairwise tests:

**44 / 45** pairs reject  $H_0: \text{ARI} \leq 0.5$

Only **KNN–SoftImpute** (0.509) borderline.

### Top ARI pairs:

- EM–DAE: 0.880
- DAE–VAE: 0.863
- EM–VAE: 0.828

### Instability correlation:

$r = 0.474$  (per-assessment vs. missing fraction)

**Takeaway:** the severity tiers survive every imputation choice; MICE is a safe default.

### H3: Clinical Unit Association — Supported

- $\chi^2 = 911.3$ ,  $p \approx 4 \times 10^{-156}$
- Cramér's  $V = 0.162$
- Rubin-pooled ( $M=5$  MICE):  $V = 0.154$  [0.049, 0.259]
- **CMH test:** association **not confounded** by diagnosis group

#### Interpretation

- Small effect size (Cohen), but well above the 0.10 threshold
- Association **holds after stratifying by diagnosis** — not a confound
- Severity tier adds explanatory power beyond diagnosis; supports severity-informed routing

## H4: Domain vs. Variable Level — Supported

**Rule:** Domain wins if it beats variable-level on the three quality metrics:  
higher Silhouette, lower mean VIF, lower condition number.

| Metric       | Domain       | Variable | Winner |
|--------------|--------------|----------|--------|
| Silhouette   | <b>0.481</b> | 0.473    | Domain |
| Mean VIF     | <b>1.88</b>  | 2.77     | Domain |
| Cond. number | <b>11.8</b>  | 61.7     | Domain |

Domain wins **all three** metrics.

**Why domain-level wins:**

1. **Comparable separation:** silhouette 0.481 vs 0.473
2. **Less multicollinearity:** VIF  $2.77 \rightarrow 1.88$ , condition number  $61.7 \rightarrow 11.8$
3. **More interpretable:** clinicians think in domains, not individual tests

## Hypothesis Summary

| Hypothesis                    | Verdict   | Key Evidence                                      |
|-------------------------------|-----------|---------------------------------------------------|
| H1: Discrete, stable clusters | Supported | 10/10 methods pass sil/noise; 3 severity tiers    |
| H2: Imputation robustness     | Supported | ARI = 0.710, 44/45 Holm-significant               |
| H3: Clinical unit assoc.      | Supported | $\chi^2 = 911$ , $V = 0.162$ , CMH not confounded |
| H4: Domain vs. variable       | Supported | Domain wins all 3 metrics (sil, VIF, cond.)       |



# Robustness Additions

## A-3: Bootstrap 95% CIs

- H2 mean ARI: 0.710 [0.704, 0.716]
- H3 Cramér's  $V$ : 0.16 [0.15, 0.18]
- H4 silhouette gap: 0.05 [0.04, 0.06]

## A-2: Holm for 45 H2 pairs

- **44 / 45** pairs reject  $\text{ARI} \leq 0.5$
- Only KNN-SoftImpute borderline

## A-4: Rubin-pooled ( $M=5$ MICE)

- Cramér's  $V = 0.154$  [0.049, 0.259]
- H4 gap = 0.050 [0.043, 0.058]

## A-5: Listwise-deletion baseline

- Complete cases: 9,245 / 17,406
- **2 clusters** (vs 3 in MICE)
- Silhouette 0.257 (vs 0.526)
- $\text{ARI vs MICE} = 0.625$

⇒ Imputation materially changes structure

## D-1: Outcome-adjacent (TSI + within-patient)

- TSI by tier: 214 / 188 / 172 d (Kruskal-Wallis  $p=4e-28$ )
- **50.2%** of multi-assessment patients stay in same tier
- Transitions move *up* the gradient (recovery) 3:1

## Key Findings & Interpretation

1. **A cognitive severity gradient, not dissociated phenotypes:** 3 tiers (Above-Average / Near-Normal / Global Impairment); PC1 = 56% of variance, all domains co-vary
2. **No attention-specific dissociation** — PC1 (56%) loads near-equally on all six domains; the tiers differ in overall level, not profile shape
3. **Imputation matters:** listwise deletion finds 2 clusters (sil 0.257) vs. 3 with MICE (sil 0.53)
4. **H2 survives correction:** 44 of 45 pairs pass Holm; every pair exceeds ARI 0.50
5. **Severity tier is not diagnosis:** CMH not confounded; adds independent explanatory power
6. **Cautionary tale:** the apparent attention phenotype was a preprocessing artefact (a mis-scaled demographic) — it dissolved under correction

# Clinical Implications

## Match intensity to severity:

- Above-Average → vocational rehab, community return
- Global Impairment → intensive, structured rehabilitation

## Decision support:

- CogDash dashboard classifies new patients
- Shows confidence for each assignment

## Method recommendations:

- Use MICE as default for clinical neuropsych data
- Test sensitivity with  $\geq 2$  imputation methods
- Prefer domain-level features for clustering
- Report stability/confidence alongside phenotype labels

# Interactive Dashboard: CogDash

## 8 modules:

1. Home — summary metrics & hypothesis verdicts
2. Cluster Explorer — t-SNE 2D & UMAP 3D
3. Cognitive Profiles — radar charts & heatmaps
4. Patient Lookup — individual patient profiles
5. Imputation Comparison — ARI/NMI matrices
6. Clinical Units — cross-tabulations
7. New Patient Classification — decision support
8. Advanced Analytics — feature importance, stability

## Tech stack:

- Streamlit + Plotly interactive plots
- Colour-blind safe palette
- Docker containerised
- Pre-computed results (pickle)

## Limitations & Mitigations

1. **Single-centre** — but tests (TMT, Stroop, digit span) are **internationally standard** for 30+ years
2. **A continuum, not natural kinds** — the tiers discretise a single severity axis; tier count (3–5) depends on resolution
3. **Feature-engineering sensitivity** — the earlier attention phenotype was an artefact of a mis-scaled demographic; corrected by winsorise + sign-align + standardise
4. **MAR assumption** — but 10-method comparison + Rubin pooling + listwise baseline act as built-in sensitivity analyses
5. **No functional outcome variables** — D-1 is only outcome-adjacent (TSI, within-patient stability)
6. **Cross-sectional** — but multiple assessments per patient give coverage of the recovery continuum (recovery direction confirmed)

## Future Directions

- **Longitudinal phenotyping** — track cognitive change over recovery
- **Multi-centre validation** — test reproducibility across settings
- **Recovery prediction** — use phenotype as a prognostic tool
- **Advanced imputation** — GANs (GAIN), transformers for larger batteries
- **Neuroimaging integration** — link phenotypes to brain structure/function

## Conclusions

1. **A cognitive severity gradient in ABI:** 3 tiers (Above-Average / Near-Normal / Global Impairment) on a single severity axis (PC1 56%), cutting across diagnostic categories
2. **Imputation is not cosmetic:** listwise deletion recovers 2 clusters (sil 0.257) vs. 3 with MICE (sil 0.53); ARI 0.625 between the two
3. **Robust across imputation:** 44 of 45 H2 pairs survive Holm; every pair exceeds ARI 0.50
4. **Domain-level features beat** variable-level on all three quality metrics (silhouette, VIF, condition number), confirmed via Rubin pooling
5. **Severity tier adds clinical value** beyond diagnosis (CMH not confounded); 86

## Methodological Contribution

An imputation-robustness framework for clustering, *plus* a worked cautionary case: preprocessing artefacts can masquerade as cognitive phenotypes unless features are scaled and screened with care.

# Thank you

Questions?

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Code & dashboard: `https://github.com/zkunos/thesis\_project`